

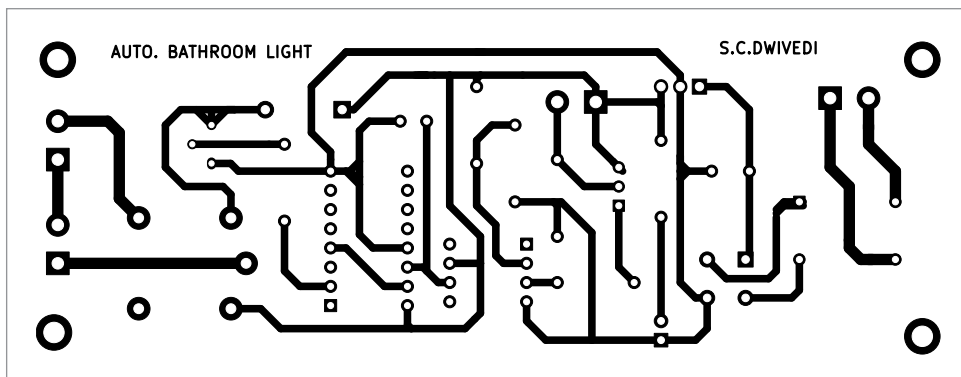


Fig. 3: Actual-size PCB of automatic bathroom light circuit

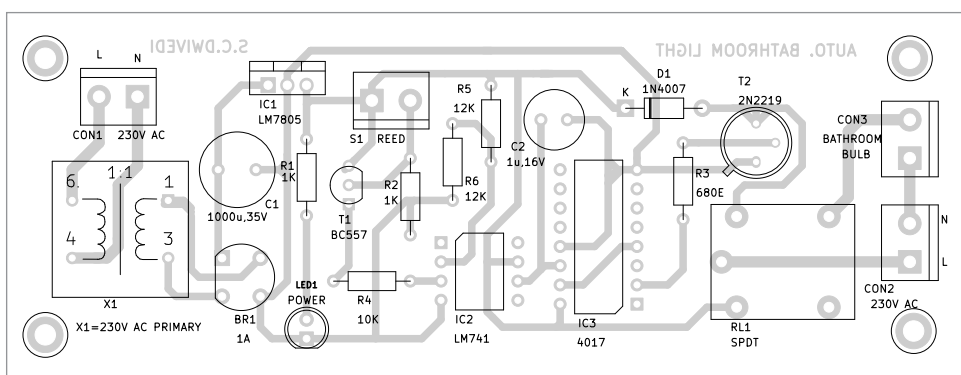


Fig. 4: Component layout of the PCB

tor T2 conducts. As a result, relay RL1 energises and bathroom light gets connected to 230V AC and glows.

PARTS LIST

Semiconductors:

IC1	- LM7805, 5V voltage regulator
IC2	- Op-amp 741
IC3	- 4017 decade counter
BR1	- 1A bridge rectifier
LED1	- 5mm LED
T1	- BC557 PNP transistor
T2	- 2N2219 NPN transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R2	- 1-kilo-ohm
R3	- 680-ohm
R4	- 10-kilo-ohm
R5, R6	- 12-kilo-ohm

Capacitors:

C1	- 1000 μ F, 35V electrolytic
C2	- 1 μ F, 16V electrolytic

Miscellaneous:

CON1-CON3	- 2-pin connector
S1	- Small normally closed type glass reed switch
RL1	- 5V, 1CO relay
X1	- 230V AC primary to 9V, 500mA secondary transformer
	- A small magnet

After entering the bathroom, when you close the door again, the magnet and the reed switch again come near each other. Reed switch contacts open and transistor T1 does not conduct. Hence voltage at pin 3 of IC2 goes lower than reference voltage at pin 2, output of op-amp IC2 at its pin 1 goes from high to low, and decade counter 4017 (IC3) does not get any clock signal. Clock Q1 of 4017 does not shift to Q2 output, the relay remains energised, and the bathroom light keeps glowing.

When you open the bathroom door once again to exit, the magnet and the reed switch move away from each other, the reed switch contacts close, and transistor T1 conducts. Hence voltage at pin 3 of IC2 goes higher than the reference voltage at pin 2, output of op-amp IC2 at its pin 1 goes from low to high, and decade counter 4017 (IC3) gets a clock signal. Clock Q1 of 4017 now

shifts to Q2 output and relay driver transistor T2 stops conducting. As a result, relay RL1 de-energises, and the bathroom light disconnects from 230V AC and goes off.

In brief, when you open the bathroom door, its light will switch on. After entering, when you close the door, the light will remain on. After completing your task in the bathroom, when you open the door to come out, the light will switch off. The light will remain off until someone wants to enter the bathroom again.

Construction and installation

An actual-size, single-side PCB layout for the automatic bathroom light circuit is shown in Fig. 3 and its component layout in Fig. 4.

After assembling the circuit on PCB, enclose it in a suitable box. Fix LED1 and CON2 in front side of the box and the bathroom bulb on the rear side of the box.

The heart of this circuit is the combination of magnet and normally closed type of reed switch, which are installed opposite to each other in the bathroom's door frame and the door, respectively. So, when the door is closed, the reed switch contacts open and when the door is opened, its contacts close. After proper installation of the magnet and the reed switch, connect them to the circuit assembled by you. Now the automatic bathroom light is ready to use. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-english-make-automatic-bathroom-light>

S.C. Dwivedi is an electronics enthusiast and circuit designer at EFY